

Cookicutter

cookiecutter -c v1 <http://github.com/drivendata/cookiecutter-data-science>

We create venv in cookiecutter template

python -m venv venv

venv\Scripts\activate

Now install requirement.txt file run

pip install -r requirements.txt

pip freeze > requirements.txt →added libraries version

Now install git and dvc

git init

dvc init

Now add some file in .gitignore

My environment in case my →venv

models

reports

Now you can add github repository

Now we add two file

params.yaml

dvc.yaml→not need to make this file because when make the pipeline then automatical make this file generate

How to make pipeline dvc

dvc stage add -n <stage_name> -d <dependency> -o <output> "<command>"

E.g

- dvc stage add -n data_ingestion -d src/data_ingestion.py -o data/raw python src/data_ingestion.py
- dvc stage add -n preprocess -d src/preprocess.py -d data/raw/customer.csv -o data/processed.csv python src/preprocess.py

How to run

dvc repro

How pipeline visualize

dvc dag

Git and dvc

Note:

Both work same let say first command write gitinit and then write dvc not add git url after dvc init add then git url and then dvc url and so on

git init	dvc init
git remote url →when I create repository then this url there	dvc remote add mytempremote C:\Users\dell\AppData\Local\Temp\abubakar-storage Or

	dvc remote add -d mytempremote s3://twitter-commit-analysis-data-001
Git commit -m 'init commit with dvc config file'	
Run code file	
git status	
	dvc add data/customer.csv
Git status Git add .	
Git commit -m 'first experiment'	
	Dvc push or dvc push -r mytempremote
Git push origin main	
Now you can may be second time change code and then may be our data ho ja change according to project need	
lets say we taka as a example second experiment	
Some code change and then my csv file will be change	
Git status	
	Dvc status
	Dvc add data/customer.csv
Git status	
Git add .	
Git commit -m 'seocnd experiment'	
	Dvc push or dvc push -r mytempremote
Git push origin main	
Git log - - oneline	
Now you receive some comment and with hash	
Now you want move then simple copy hash let's say heash 234f564 this for first experiment	

Git checkout 234f564	
	Dvc checkout

Note:

Now if your team member clone the repository then simple command for the data

Dvc featch

Dvc pull

In pile line case

Dvc fetch

Dvc checkout →then data take according to code version

Data versioning pipeline and on aws using dvc

For pipeline

Step 1 - Set up cookiecutter -> delete file→ renames models to model in src folder

Step 2 - Initialize git

Step 3 - Initialize DVC

Step 4 - Create a GitHub repository

Step 5 - Add remote for git

Step 6 - Add remote for DVC

Step 7 - Git add -> First commit

Step 8 - Add data ingestion -> Add data preprocessing -> Add feature engg -> Add model training -> Add model eval

Step 9 - Add dvc.yaml -> Add params.yaml

Step 10 - DVC repro

Step 11 - git status / dvc status

Step 12 - Git add code/dvc files→here in pipeline automatically add dvc file I mean data but we perform add manually code file for git

Step 13 - Git commit

Step 14 - DVC push to remote(using this command dvc push -r mytempremote)

Step 15 - Git push to GitHub

Note:

Now if other person clone the my repository then there don't have data so He write some command

Dvc fetch

Dvc checkout

For AWS S3

```
File Edit View

Create S3 bucket

Create IAM user

pip install dvc[s3]

pip install awscli

git init

dvc init

git remote add
aws configure
dvc remote add -d dvcstore s3://campusxdvcbucket
dvc add
git commit
dvc push
git push
```



How to connect s3 bucket and laptop

- I know How make s3
 - **Now How make IAM user?**
 - Just search IAM and then open the page and then select **User** in Access management
 - And then click button by this name **create user** then open the new page and there page enter user name you choose any name and then if you give access management then you can otherwise if you only access for s3 then we move next step
 - Step 2 set permission→if only select s3 then my user only access my s3 bucket not any other things access but I don't have s3 bucket policies so now we make this policies
 - In step 2 select if i have s3 polices then attach policies otherwise we make new polices so we click on sidebar access management and then click on policies and then click the create policy
 - Then you will see **policy editor** so click on json button and then enter this code in json file And we will some changing in code like we go to s3 bucket and then go to properties and then copy **Amazon Resource Name (ARN)** and then edit the code in both Resource key
- ```
{
 "Version": "2012-10-17",
 "Statement": [
 {
```

```

 "Effect": "Allow",
 "Action": ["s3:ListBucket"],
 "Resource": "arn:aws:s3:::twcommitanalysis12"
 },
 {
 "Effect": "Allow",
 "Action": ["s3:PutObject", "s3:GetObject"],
 "Resource": "arn:aws:s3:::twcommitanalysis12/*"
 }
]
}

```

- Then finally click next and then will open new page
- So now enter the police details like police name and description and you can give any name for police name
- Then click on create police
- Then now we go to policies page and then select his policies i mean we go to set permissions wala page pa and then select the permission option in select this **Attach policies directly** aur waha ja kar search kar ga name aur select kar ga jo ham na polic bania ha
- And then next and then check the name user and policy for confirmation and then click on **create user button**
- And then open the user and then we make the key
- So click on the **create access key** and then will open the new page and then select **Command Line Interface (CLI)** and then check the Confirmation and then next and then without changing click on the **create access key button**
- **And then you can copy**

Access key

AKIA5GLVD

EE6TB6

COX4A

Secret access key

PHgSMFVTgy6ozRa35ytBhM7HnT79iHJtb7PVCruM

- 
- Or download the csv but careful these key ham axch tari samlaa kar rakni ha
  - Then we move on project and perform these steps
  - Git init
  - Dvc init
  - Git add remote
  - Befour dvc add remote install some libraries such as
  - Pip install dvc[s3]
  - Pip install awscli
  - Aws configure → just this command write and when write this command then aws ask me like
  - AWS access key id :AKIA5GLVDEE6TB6COX4A
  - AWS secret key :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
  - Default region name [ap-southeast-2]:if this was already selected in aws then okay otherwise we enter the region which jo ham na aws pa select kia hoe ha
  - Default output format[json]: so we do just enter

- And then finally we add `dvc remote add -d allowdvc s3://twcommitanalysis12`
- Git commit
- Dvc push
- Git push
- Reference: <https://ritza.co/articles/dvc-s3-set-up-s3-as-dvc-remote/>

## Amazon practical on EC2,S3,IAM

### How make EC2?

- Click EC2 and then click instance and then provide name in **Name and tags box** and then click ubuntu but ensure free tier eligible
- Check instance type there already select **t2.micro** but ensure this t2.micro is free tier eligible so you don't need any changing if select t2.micro
- Generate key pair (login):click **Create new key pair** then enter key pair name and key pair type select RSA and Private key file format select .pem  
Then download the one file but that file is very very very important because If that file access other person then that person access my ec2 so we do store in safe place or hidden the file in safe place
- In Network setting then check box **Allow SSH traffic from** and then click on the **edit** button this button place on network setting box top right side and then you do add one more add security group rule and then just change port range 8501 this digit for i think only streamlit app  
*note:And in youtube comment analysis for sir select three checkbox and then i don't select edit button and not need in details in edit button like upper this*  
Then check configure storage and select only 8gb but I have 30 gb and you can select more gb but in under the 30 gb  
Then click launch instance button and then automatically generate the EC2
- After generate the EC2 then you click on the instance id and then click on the connect button and then open the new page that page name connect to instance but not any changing just scroll down and click the connect button and then open the consol on the screen now you do just install the libraries according to project require  
But we put some command before libraries install  

```

sudo apt-get update
sudo apt-get install python3 python3-venv python3-pip -y
python3 -m venv mlflow_env→create the venv in case my ven name mlflow_env
source mlflow_env/bin/activate→activate the venv
Libraries install
pip install mlflow boto3
pip list

```

## S3 bucket

- Just search s3 and then open the page and that page on button click by this name **create bucket** then you give the name but that name is unique in the whole world

- Then not change on the whole page just scroll down and then click the button create bucket so after click button our s3 bucket created

## **IAM→this IAM goal my EC2 communicate with s3**

- Just search IAM and then open the page and then select **Role** in Access management
- Then click the button that button name **create role**
- There have 3 steps we do in first step select **Aws service** and **Service or use case** select for **EC2**
- In steps 2 you select policies by this name **AmazonS3fullAccess**
- In step 3 you give the role name and then not any changing just scroll down and click the button create role

### **IAM Role update**

- Just go back EC2 page and then select **action button** on the top right side and in action button you select **security** and then select **Modify IAM role**
- Then open the new page and there is select IMA Role by that name jo ap na bana ha but In my case ytanalysis\_role and then just click on update button

### **MLflow run on EC2 with screen libraries**

#### **Just these command run on ec2 console**

- `sudo apt-get install screen -y`
- `screen -S mlflow`
- `mlflow server --backend-store-uri ./mlruns --default-artifact-root s3://your-s3-bucket-name --host 0.0.0.0 --port 5000` → jaha par your-s3-bucket-name ha jaha ap na apna bucket ka name dana ha aur ja name jaha bucket bana ta waha hota ha
- Then after these command open the 5000 port on the ec2 so we go to EC2 page **i-0e7a930ff4f1269e5 (ml\_prectice)** then select **security tab** and then click **security groups** then open the new page and select **inbound rule tab** and then in tab select right side button **Edit inbound rule**
- After edit inbound rule button open the new page then click on the **add rule button** and then select the port range 5000 and ip 0.0.0.0/0
- Then just scroll down and save rule

# Steps to setup a MLflow Tracking Server

## Steps:

1. Create EC2 instance
2. `sudo apt-get update`
3. `sudo apt-get install python3 python3-venv python3-pip -y`
4. `python3 -m venv mlflow_env`
5. `source mlflow_env/bin/activate`
6. `pip install mlflow boto3`
7. `pip list`
8. Create S3 bucket
9. Setup IAM Role → AmazonS3FullAccess → EC2-MLflow-S3-Access
10. `sudo apt-get install screen -y`
11. `screen -S mlflow`
12. `mlflow server --backend-store-uri ./mlruns  
--default-artifact-root s3://your-s3-bucket-name --host  
0.0.0.0 --port 5000`
13. Setup security group

## If I have already make all things then open mlflow ui use this root after close aws

EC2->instance->click the instance→click connect button→then open the new page not any changing just click connect button and then open the screen like cmd type

So enter this command

In this command enter the apna s3 bucket ka name

```
mlflow server --backend-store-uri ./mlruns --default-artifact-root
s3://your-s3-bucket-name --host
```

Then one step more

EC2->instance->Public DNS

Like this [ec2-18-234-99-113.compute-1.amazonaws.com](https://ec2-18-234-99-113.compute-1.amazonaws.com)

And then write 5000 like this

[ec2-18-234-99-113.compute-1.amazonaws.com:5000](https://ec2-18-234-99-113.compute-1.amazonaws.com:5000) and then enter

My reddit project key user role

Access key = AKIA5GLVDEE65GAKJ5PH



Secret key = wADowh7+P+kpGCXBj3vUacZ9Sjfj1VfN5eTB4HVv

# Docker

- If I have image then **Pull the image** and below command run
- **These command use in docker terminal**
- Docker pull saddiquetool/lappy → saddiquetool is a username and lappy is a image
- Docker run -p 8501:8501 saddiquetool/lappy → run the image

**Now I am going to make docker image**

- **Step1**
- **In vscode make a Dockerfile**  
**E.g dockerfile**  
# Use official Python image as base  
FROM python:3.12

# Set working directory inside container  
WORKDIR /app

# Copy requirements.txt and install dependencies  
COPY ./app  
RUN pip install -r requirements.txt

# environment  
EXPOSE=5000

**Note: for flask keep 5000 and for streamlit 8501**

# Command to run the app  
CMD ["python", "app.py"]

- After these process go to terminal in vscode and run this command
- **Docker image create command below**
- Docker build -t saddiquetool/table . → table is a image name you can keep any name and dot represent where is build image we just write after image name dot (.) then just enter and dot represent current folder
- After complete process then you can see the image in docker desktop software in image tab
- **Docker image run use below command after image created completed**
- Docker run -p 8888:5000 saddiquetool/table:latest
- **Explanation:** 8888 show my machine port and 5000 show in docker port because I use docker file so we keep both port equal using (:)
- And latest show as a tag after docker image name
- Then you can see in docker desktop in container tab
- **Now we image push on the docker hub**

- Docker login
- Docker push saddiquetool/table:latest

## Some question

I want specific folder make a requirements.txt with all libareies jo require ha aso folder ma code ka lia such as we will use for flask\_app folder

- Install this libraries
- pip install pipreqs
- Go to that folder location like cd flask\_app
- pipreqs . - - force

We use gunicorn server rather then flask server

- Because flask servers at a time only one request handle but gunicorn servers many request handle at a time

## Deployments methods

First method

### Step for model to dockerization to EC2

Already know how to make docker file for model and Ec2 now only check how to dockerfile deploy on ec2

- create an ec2 instance
- Then open ec2 and connect then will open terminal then these below command write
- sudo apt-get update
- sudo apt-get install -y docker.io
- sudo systemctl start docker
- sudo systemctl enable docker
- Sudo usermod -aG docker \$user → user mean your ec2 user like ubuntu
- Add secret key on github  
EC2\_HOST → open ec2 and find public ipv4 Dns so just copy and added in github secret variable  
EC2\_USER → find user name like ubuntu  
EC2\_SSH\_KEY → when I made ec2 then I downloaded pem file so just open the file and as it copy all content and pates on github
- Modify ci.yaml  
- name: Deploy to EC2

```

if: success()
uses: appleboy/ssh-action@v0.1.5
with:
 host: ${ secrets.EC2_HOST }
 username: ${ secrets.EC2_USER }
 key: ${ secrets.EC2_SSH_KEY }
 script: |
 docker pull ${ secrets.DOCKER_HUB_USERNAME }/emotion2:latest
 →replace with your image name and tag
 docker stop my-app || true
 docker rm my-app || true
 docker run -p 80:5000 --name my-app \
 -e DAGSHUB_PAT=${ secrets.DAGSHUB_PAT } \
 ${ secrets.DOCKER_HUB_USERNAME }/emotion2:latest →dagshub path
 and docker hub username add your and ana ko be ap na github secret variable
 ma add karna ha

```

- Ci file push on github
- AWS security group  
Open ec2 and then scroll down and go to security tab and then there will make inbound wala seen

## Second method

### Step for model to dockerization and ECR→ docker image push on ECR not docker hub

#### ECR

Search **Elastic container registry** then click it and then will open the new page and then click on create button then again will open the new page then you give the name repository there and you can choose any name then not any changing on the whole page just go down and then click on the **create** button

#### FOR manually

##### Push the image on ECR

Then select the repository and click on the button **view push command** then show all command and then just one by one enter in vscode after set the aws configure

**Note:** jaha jaha pa hybrid write kya hoe ja repo ka name jab ap repo bana goo jasi name sa be wo waha pa show ho jaa gaii command ma he so no tension

Command like this

- `aws ecr get-login-password --region eu-north-1 | docker login --username AWS --password-stdin 331374384470.dkr.ecr.eu-north-1.amazonaws.com`

- `docker build -t hybird .`
- `docker tag hybird:latest`  
[331374384470.dkr.ecr.eu-north-1.amazonaws.com/hybird:latest](https://331374384470.dkr.ecr.eu-north-1.amazonaws.com/hybird:latest)
- `docker push 331374384470.dkr.ecr.eu-north-1.amazonaws.com/hybird:latest`

**If you want Pull the image from ECR to machine then these below commands used**

- `aws ecr get-login-password --region eu-north-1 | docker login --username AWS --password-stdin` [331374384470.dkr.ecr.eu-north-1.amazonaws.com](https://331374384470.dkr.ecr.eu-north-1.amazonaws.com)
- `docker pull 331374384470.dkr.ecr.eu-north-1.amazonaws.com/hybird:latest`
- `Docker run -d -p 8888:5000 -e DAGSHUB_PAT=e691c7193ab345ec4245322x3242 --name campusx-app`  
[992382647488.dkr.ecr.ap-southeast-2.amazonaws.com/campusx\\_ecr:v1](https://992382647488.dkr.ecr.ap-southeast-2.amazonaws.com/campusx_ecr:v1) → according to your names this command change so check it when you will make a project any

**How to docker image deploy from ECR to Ec2**

- Create ec2 machine
- Then connect and these command runs below
- `sudo apt-get update`
- `sudo apt-get install -y docker.io`
- `sudo systemctl start docker`
- `sudo systemctl enable docker`
- `sudo apt-get update`
- `sudo apt-get install -y unzip curl`
- `curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"`
- `unzip awscliv2.zip`
- `sudo ./aws/install`
- `sudo usermod -aG docker ubuntu`
- `aws configure` → told there your access key and secret key
- `aws ecr get-login-password --region ap-southeast-2 | docker login --username AWS --password-stdin` [992382647488.dkr.ecr.ap-southeast-2.amazonaws.com](https://992382647488.dkr.ecr.ap-southeast-2.amazonaws.com)
- `docker pull`  
[992382647488.dkr.ecr.ap-southeast-2.amazonaws.com/campusx\\_ecr:v1](https://992382647488.dkr.ecr.ap-southeast-2.amazonaws.com/campusx_ecr:v1) → this command will change according to your project repo so i think this command copy ahead from ecr repo command
- `docker run -d -p 80:5000 \`  
`-e DAGSHUB_PAT=e691c7193ab61dc9678e31c6b92ded8a65f80697 \`  
`--name campusx-app \`  
[992382647488.dkr.ecr.ap-southeast-2.amazonaws.com/campusx\\_ecr:v1](https://992382647488.dkr.ecr.ap-southeast-2.amazonaws.com/campusx_ecr:v1) → if you don't use mlflow then not need dagshub part just remove dagshub token and other key will be same but we mostly use as a data scientist but in my hybrid recommend system we are not using
- Now after completed you go to ec2 and open the security tab and edit the inbound http with 0/0/0/0

## FOR Ci/Cd

- First added aws access and secret key on github secret variable
- Then below code added in ci.yaml file and according to change region and docker image name etc according to your project and most command manually wali use ki ha jo hum na jab ecr ki repo bani taii wo wali commands
- **Push image on ECR**
- - name: Login to AWS ECR  
run: |  
aws configure set aws\_access\_key\_id \${ secrets.AWS\_ACCESS\_KEY\_ID }  
aws configure set aws\_secret\_access\_key \${ secrets.AWS\_SECRET\_ACCESS\_KEY }  
}}  
aws ecr get-login-password --region ap-southeast-2 | docker login --username AWS --password-stdin \992382647488.dkr.ecr.ap-southeast-2.amazonaws.com
- name: Build Docker image  
run: |  
docker build -t campusx\_ecr .
- name: Tag Docker image  
run: |  
docker tag campusx\_ecr:latest \992382647488.dkr.ecr.ap-southeast-2.amazonaws.com/campusx\_ecr:latest
- name: Push Docker image to AWS ECR  
run: |  
docker push \  
[992382647488.dkr.ecr.ap-southeast-2.amazonaws.com/campusx\\_ecr:latest](https://992382647488.dkr.ecr.ap-southeast-2.amazonaws.com/campusx_ecr:latest)

### Push image ECR to EC2

- - create ec2
- Run these command on ec2 terminal after connect button click
- sudo apt-get update
- sudo apt-get install -y docker.io
- sudo systemctl start docker
- sudo systemctl enable docker
- sudo apt-get install -y unzip curl
- curl "https://awscli.amazonaws.com/awscli-exe-linux-x86\_64.zip" -o "awscliv2.zip"
- unzip awscliv2.zip
- sudo ./aws/install
- sudo usermod -aG docker ubuntu
- exit
- check docker --version
- - add pem file as secret key to GitHub secrets
- - name: Deploy to EC2

```

if: success()
uses: appleboy/ssh-action@v0.1.5
with:
 host: ${ secrets.EC2_HOST }
 username: ${ secrets.EC2_USER }
 key: ${ secrets.EC2_SSH_KEY }
 script: |
 aws configure set aws_access_key_id ${ secrets.AWS_ACCESS_KEY_ID }
 aws configure set aws_secret_access_key ${ secrets.AWS_SECRET_ACCESS_KEY }
 aws configure set default.region ap-southeast-2
 aws ecr get-login-password --region ap-southeast-2 | docker login --username AWS
 --password-stdin \ 992382647488.dkr.ecr.ap-southeast-2.amazonaws.com
 docker pull 992382647488.dkr.ecr.ap-southeast-2.amazonaws.com/campusx_ecr:latest
 docker stop my-app || true
 docker rm my-app || true
 docker run -d -p 80:5000 -e DAGSHUB_PAT=${ secrets.DAGSHUB_PAT } \
 --name campusx-app
992382647488.dkr.ecr.ap-southeast-2.amazonaws.com/campusx_ecr:latest

```

- Then save ci.yaml file and push the code on github
- Add security group in Ec2 like inbound ma ja kar http

**note:**

## **Project steps common**

1. Make a all code in evaluation and registry use mlflow if you want use mlflow
2. Dvc pipeline
3. Data push on s3 bucket using dvc
4. Docker image create
5. Docker image push on ec2